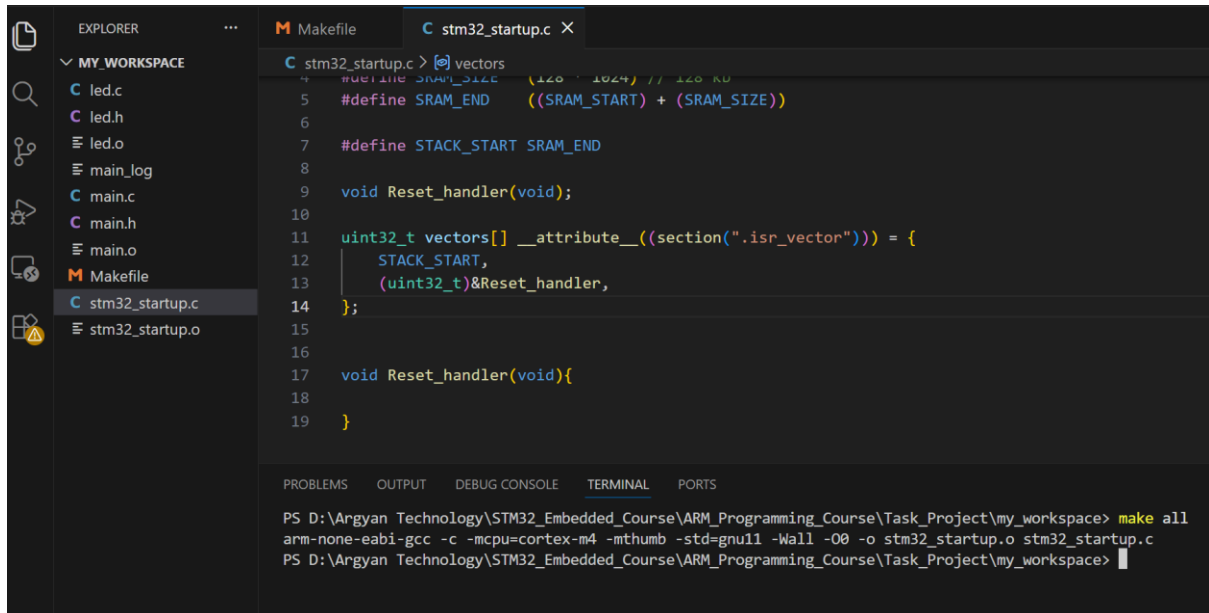


Startup File Exercise

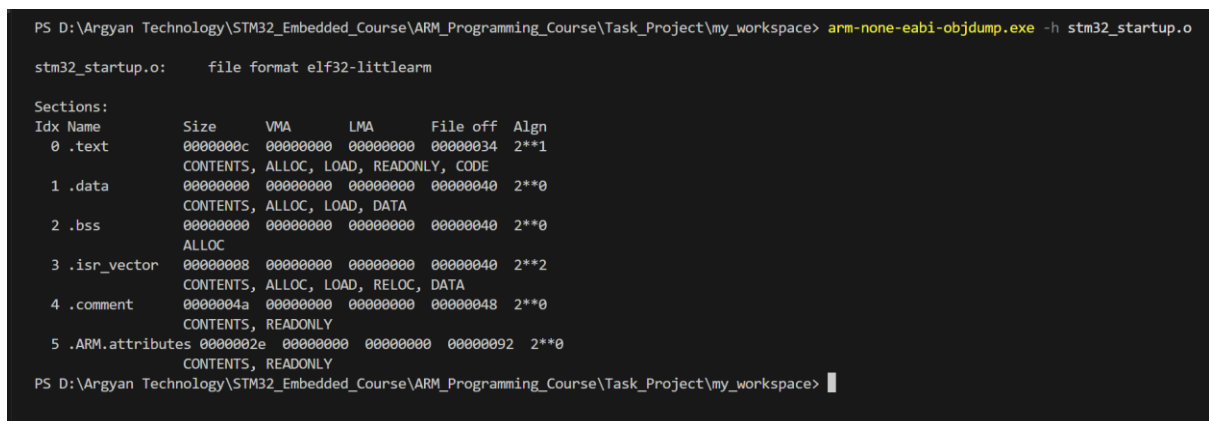


The screenshot shows the Visual Studio Code interface. The Explorer pane on the left displays the project structure under 'MY_WORKSPACE', including files like led.c, led.h, led.o, main_log, main.c, main.h, main.o, Makefile, stm32_startup.c, and stm32_startup.o. The main editor window shows the contents of 'stm32_startup.c'. The code defines SRAM_SIZE, SRAM_START, and SRAM_END, and sets up the .isr_vector section with a Reset_handler function. The terminal at the bottom shows the execution of 'make all' using the arm-none-eabi-gcc compiler, resulting in the creation of stm32_startup.o.

```
1 #include <stdint.h>
2 #define SRAM_SIZE 0x1000 // 4KB
3
4 #define SRAM_START 0x20000000
5 #define SRAM_END ((SRAM_START) + (SRAM_SIZE))
6
7 #define STACK_START SRAM_END
8
9 void Reset_handler(void);
10
11 uint32_t vectors[] __attribute__((section(".isr_vector"))) = {
12     STACK_START,
13     (uint32_t)&Reset_handler,
14 };
15
16
17 void Reset_handler(void){
18
19 }
```

```
PS D:\Argyan Technology\STM32_Embedded_Course\ARM_Programming_Course\Task_Project\my_workspace> make all
arm-none-eabi-gcc -c -mcpu=cortex-m4 -mthumb -std=gnu11 -Wall -O0 -o stm32_startup.o stm32_startup.c
PS D:\Argyan Technology\STM32_Embedded_Course\ARM_Programming_Course\Task_Project\my_workspace>
```

Startup File Sections



The screenshot shows a terminal window with the command 'arm-none-eabi-objdump.exe -h stm32_startup.o' executed. The output displays the file format as 'elf32-littlearm' and lists the sections of the object file. The sections table includes indices, names, sizes, virtual addresses (VMA), load addresses (LMA), file offsets, and alignments.

```
PS D:\Argyan Technology\STM32_Embedded_Course\ARM_Programming_Course\Task_Project\my_workspace> arm-none-eabi-objdump.exe -h stm32_startup.o
stm32_startup.o:      file format elf32-littlearm

Sections:
Idx Name          Size      VMA       LMA       File off  Algn
 0 .text          0000000c  00000000  00000000  00000034  2**1
   CONTENTS, ALLOC, LOAD, READONLY, CODE
 1 .data          00000000  00000000  00000000  00000040  2**0
   CONTENTS, ALLOC, LOAD, DATA
 2 .bss           00000000  00000000  00000000  00000040  2**0
   ALLOC
 3 .isr_vector     00000008  00000000  00000000  00000040  2**2
   CONTENTS, ALLOC, LOAD, RELOC, DATA
 4 .comment        0000004a  00000000  00000000  00000048  2**0
   CONTENTS, READONLY
 5 .ARM.attributes 0000002e  00000000  00000000  00000092  2**0
   CONTENTS, READONLY
PS D:\Argyan Technology\STM32_Embedded_Course\ARM_Programming_Course\Task_Project\my_workspace>
```

Random Section Reset_handler

```
Sections:
Idx Name          Size      VMA       LMA       File off  Align
 0 .text          00000000 00000000 00000000 00000034 2**1
                  CONTENTS, ALLOC, LOAD, READONLY, CODE
 1 .data          00000000 00000000 00000000 00000034 2**0
                  CONTENTS, ALLOC, LOAD, DATA
 2 .bss           00000000 00000000 00000000 00000034 2**0
                  ALLOC
 3 .random_section 0000000c 00000000 00000000 00000034 2**1
                  CONTENTS, ALLOC, LOAD, READONLY, CODE
 4 .isr_vector     00000008 00000000 00000000 00000040 2**2
                  CONTENTS, ALLOC, LOAD, RELOC, DATA
 5 .comment        0000004a 00000000 00000000 00000048 2**0
                  CONTENTS, READONLY
 5 .comment        0000004a 00000000 00000000 00000048 2**0
 5 .comment        0000004a 00000000 00000000 00000048 2**0
 5 .comment        0000004a 00000000 00000000 00000048 2**0
                  CONTENTS, READONLY
 6 .ARM.attributes 0000002e 00000000 00000000 00000092 2**0
                  CONTENTS, READONLY
PS D:\Argyan Technology\STM32_Embedded_Course\ARM_Programming_Course\Task_Project\my_workspace> arm-none-eabi-objdump.exe -d stm32_startup.o

stm32_startup.o:      file format elf32-littlearm

Disassembly of section .random_section:

00000000 <Reset_handler>:
 0: b480      push    {r7}
 2: af00      add     r7, sp, #0
 4: bf00      nop
 6: 46bd      mov     sp, r7
 8: bc80      pop     {r7}
 a: 4770      bx     lr
```

Startup File Code

```
#include <stdint.h>

#define SRAM_START 0x20000000
#define SRAM_SIZE (128 * 1024) // 128 kb
#define SRAM_END ((SRAM_START) + (SRAM_SIZE))

#define STACK_START SRAM_END

/* function prototypes of STM32F407x system exception and IRQ handlers */

void Reset_Handler(void);

void NMI_Handler(void) __attribute__((weak, alias("Default_Handler")));
void HardFault_Handler(void) __attribute__((weak, alias("Default_Handler")));
void MemManage_Handler(void) __attribute__((weak, alias("Default_Handler")));
void BusFault_Handler(void) __attribute__((weak, alias("Default_Handler")));
void UsageFault_Handler(void) __attribute__((weak, alias("Default_Handler")));
void SVC_Handler(void) __attribute__((weak, alias("Default_Handler")));
void DebugMon_Handler(void) __attribute__((weak, alias("Default_Handler")));
void PendSV_Handler(void) __attribute__((weak, alias("Default_Handler")));
void SysTick_Handler(void) __attribute__((weak, alias("Default_Handler")));
void WWDG_IRQHandler(void) __attribute__((weak, alias("Default_Handler")));
void PVD_IRQHandler(void) __attribute__((weak, alias("Default_Handler")));
void TAMP_STAMP_IRQHandler(void) __attribute__((weak, alias("Default_Handler")));
void RTC_WKUP_IRQHandler(void) __attribute__((weak, alias("Default_Handler")));
void RCC_IRQHandler(void) __attribute__((weak, alias("Default_Handler")));
void EXTI0_IRQHandler(void) __attribute__((weak, alias("Default_Handler")));
void EXTI1_IRQHandler(void) __attribute__((weak, alias("Default_Handler")));
void EXTI2_IRQHandler(void) __attribute__((weak, alias("Default_Handler")));
void EXTI3_IRQHandler(void) __attribute__((weak, alias("Default_Handler")));
void EXTI4_IRQHandler(void) __attribute__((weak, alias("Default_Handler")));
void DMA1_Stream0_IRQHandler(void) __attribute__((weak, alias("Default_Handler")));
void DMA1_Stream1_IRQHandler(void) __attribute__((weak, alias("Default_Handler")));
void DMA1_Stream2_IRQHandler(void) __attribute__((weak, alias("Default_Handler")));
void DMA1_Stream3_IRQHandler(void) __attribute__((weak, alias("Default_Handler")));
void DMA1_Stream4_IRQHandler(void) __attribute__((weak, alias("Default_Handler")));
void DMA1_Stream5_IRQHandler(void) __attribute__((weak, alias("Default_Handler")));
void DMA1_Stream6_IRQHandler(void) __attribute__((weak, alias("Default_Handler")));
void ADC_IRQHandler(void) __attribute__((weak, alias("Default_Handler")));
void CAN1_TX_IRQHandler(void) __attribute__((weak, alias("Default_Handler")));
void CAN1_RX0_IRQHandler(void) __attribute__((weak, alias("Default_Handler")));
void CAN1_RX1_IRQHandler(void) __attribute__((weak, alias("Default_Handler")));
void CAN1_SCE_IRQHandler(void) __attribute__((weak, alias("Default_Handler")));
void EXTI9_5_IRQHandler(void) __attribute__((weak, alias("Default_Handler")));
void TIM1_BRK_TIM9_IRQHandler(void) __attribute__((weak, alias("Default_Handler")));
void TIM1_UP_TIM10_IRQHandler(void) __attribute__((weak, alias("Default_Handler")));
void TIM1_TRG_COM_TIM11_IRQHandler(void) __attribute__((weak, alias("Default_Handler")));
void TIM1_CC_IRQHandler(void) __attribute__((weak, alias("Default_Handler")));
void TIM2_IRQHandler(void) __attribute__((weak, alias("Default_Handler")));
void TIM3_IRQHandler(void) __attribute__((weak, alias("Default_Handler")));
void TIM4_IRQHandler(void) __attribute__((weak, alias("Default_Handler")));
void I2C1_EV_IRQHandler(void) __attribute__((weak, alias("Default_Handler")));
void I2C1_ER_IRQHandler(void) __attribute__((weak, alias("Default_Handler")));
void I2C2_EV_IRQHandler(void) __attribute__((weak, alias("Default_Handler")));
void I2C2_ER_IRQHandler(void) __attribute__((weak, alias("Default_Handler")));
void SPI1_IRQHandler(void) __attribute__((weak, alias("Default_Handler")));
void SPI2_IRQHandler(void) __attribute__((weak, alias("Default_Handler")));
void USART1_IRQHandler(void) __attribute__((weak, alias("Default_Handler")));
void USART2_IRQHandler(void) __attribute__((weak, alias("Default_Handler")));
void USART3_IRQHandler(void) __attribute__((weak, alias("Default_Handler")));
void EXTI15_10_IRQHandler(void) __attribute__((weak, alias("Default_Handler")));
void RTC_Alarm_IRQHandler(void) __attribute__((weak, alias("Default_Handler")));
void OTG_FS_WKUP_IRQHandler(void) __attribute__((weak, alias("Default_Handler")));
void TIM8_BRK_TIM12_IRQHandler(void) __attribute__((weak, alias("Default_Handler")));
void TIM8_UP_TIM13_IRQHandler(void) __attribute__((weak, alias("Default_Handler")));
void TIM8_TRG_COM_TIM14_IRQHandler(void) __attribute__((weak, alias("Default_Handler")));
void TIM8_CC_IRQHandler(void) __attribute__((weak, alias("Default_Handler")));
void DMA1_Stream7_IRQHandler(void) __attribute__((weak, alias("Default_Handler")));
void FSMC_IRQHandler(void) __attribute__((weak, alias("Default_Handler")));
void SDIO_IRQHandler(void) __attribute__((weak, alias("Default_Handler")));
void TIM5_IRQHandler(void) __attribute__((weak, alias("Default_Handler")));
void SPI3_IRQHandler(void) __attribute__((weak, alias("Default_Handler")));
void UART4_IRQHandler(void) __attribute__((weak, alias("Default_Handler")));
void UART5_IRQHandler(void) __attribute__((weak, alias("Default_Handler")));
```

```

void TIM6_DAC_IRQHandler(void) __attribute__((weak, alias("Default_Handler")));
void TIM7_IRQHandler(void) __attribute__((weak, alias("Default_Handler")));
void DMA2_Stream0_IRQHandler(void) __attribute__((weak, alias("Default_Handler")));
void DMA2_Stream1_IRQHandler(void) __attribute__((weak, alias("Default_Handler")));
void DMA2_Stream2_IRQHandler(void) __attribute__((weak, alias("Default_Handler")));
void DMA2_Stream3_IRQHandler(void) __attribute__((weak, alias("Default_Handler")));
void DMA2_Stream4_IRQHandler(void) __attribute__((weak, alias("Default_Handler")));
void ETH_IRQHandler(void) __attribute__((weak, alias("Default_Handler")));
void ETH_WKUP_IRQHandler(void) __attribute__((weak, alias("Default_Handler")));
void CAN2_TX_IRQHandler(void) __attribute__((weak, alias("Default_Handler")));
void CAN2_RX0_IRQHandler(void) __attribute__((weak, alias("Default_Handler")));
void CAN2_RX1_IRQHandler(void) __attribute__((weak, alias("Default_Handler")));
void CAN2_SCE_IRQHandler(void) __attribute__((weak, alias("Default_Handler")));
void OTG_FS_IRQHandler(void) __attribute__((weak, alias("Default_Handler")));
void DMA2_Stream5_IRQHandler(void) __attribute__((weak, alias("Default_Handler")));
void DMA2_Stream6_IRQHandler(void) __attribute__((weak, alias("Default_Handler")));
void DMA2_Stream7_IRQHandler(void) __attribute__((weak, alias("Default_Handler")));
void USART6_IRQHandler(void) __attribute__((weak, alias("Default_Handler")));
void I2C3_EV_IRQHandler(void) __attribute__((weak, alias("Default_Handler")));
void I2C3_ER_IRQHandler(void) __attribute__((weak, alias("Default_Handler")));
void OTG_HS_EP1_OUT_IRQHandler(void) __attribute__((weak, alias("Default_Handler")));
void OTG_HS_EP1_IN_IRQHandler(void) __attribute__((weak, alias("Default_Handler")));
void OTG_HS_WKUP_IRQHandler(void) __attribute__((weak, alias("Default_Handler")));
void OTG_HS_IRQHandler(void) __attribute__((weak, alias("Default_Handler")));
void DCMI_IRQHandler(void) __attribute__((weak, alias("Default_Handler")));
void CRY_P_IRQHandler(void) __attribute__((weak, alias("Default_Handler")));
void HASH_RNG_IRQHandler(void) __attribute__((weak, alias("Default_Handler")));
void FPU_IRQHandler(void) __attribute__((weak, alias("Default_Handler")));

```

```

uint32_t vectors[] __attribute__((section(".isr_vector"))) = {

```

```

    STACK_START,
    (uint32_t)&Reset_Handler,
    (uint32_t)&NMI_Handler,
    (uint32_t)&HardFault_Handler,
    (uint32_t)&MemManage_Handler,
    (uint32_t)&BusFault_Handler,
    (uint32_t)&UsageFault_Handler,
    (uint32_t)&SVC_Handler,
    (uint32_t)&DebugMon_Handler,
    (uint32_t)&PendSV_Handler,
    (uint32_t)&SysTick_Handler,
    (uint32_t)&WWDG_IRQHandler,
    (uint32_t)&PVD_IRQHandler,
    (uint32_t)&TAMP_STAMP_IRQHandler,
    (uint32_t)&RTC_WKUP_IRQHandler,
    (uint32_t)&RCC_IRQHandler,
    (uint32_t)&EXTI0_IRQHandler,
    (uint32_t)&EXTI1_IRQHandler,
    (uint32_t)&EXTI2_IRQHandler,
    (uint32_t)&EXTI3_IRQHandler,
    (uint32_t)&EXTI4_IRQHandler,
    (uint32_t)&DMA1_Stream0_IRQHandler,
    (uint32_t)&DMA1_Stream1_IRQHandler,
    (uint32_t)&DMA1_Stream2_IRQHandler,
    (uint32_t)&DMA1_Stream3_IRQHandler,
    (uint32_t)&DMA1_Stream4_IRQHandler,
    (uint32_t)&DMA1_Stream5_IRQHandler,
    (uint32_t)&DMA1_Stream6_IRQHandler,
    (uint32_t)&ADC_IRQHandler,
    (uint32_t)&CAN1_TX_IRQHandler,
    (uint32_t)&CAN1_RX0_IRQHandler,
    (uint32_t)&CAN1_RX1_IRQHandler,
    (uint32_t)&CAN1_SCE_IRQHandler,
    (uint32_t)&EXTI9_5_IRQHandler,
    (uint32_t)&TIM1_BRK_TIM9_IRQHandler,
    (uint32_t)&TIM1_UP_TIM10_IRQHandler,
    (uint32_t)&TIM1_TRG_COM_TIM11_IRQHandler,
    (uint32_t)&TIM1_CC_IRQHandler,
    (uint32_t)&TIM2_IRQHandler,
    (uint32_t)&TIM3_IRQHandler,
    (uint32_t)&TIM4_IRQHandler,
    (uint32_t)&I2C1_EV_IRQHandler,
    (uint32_t)&I2C1_ER_IRQHandler,
    (uint32_t)&I2C2_EV_IRQHandler,
    (uint32_t)&I2C2_ER_IRQHandler,
    (uint32_t)&SPI1_IRQHandler,
    (uint32_t)&SPI2_IRQHandler,

```

```

(uint32_t)&USART1_IRQHandler,
(uint32_t)&USART2_IRQHandler,
(uint32_t)&USART3_IRQHandler,
(uint32_t)&EXTI15_10_IRQHandler,
(uint32_t)&RTC_Alarm_IRQHandler,
(uint32_t)&OTG_FS_WKUP_IRQHandler,
(uint32_t)&TIM8_BRK_TIM12_IRQHandler,
(uint32_t)&TIM8_UP_TIM13_IRQHandler,
(uint32_t)&TIM8_TRG_COM_TIM14_IRQHandler,
(uint32_t)&TIM8_CC_IRQHandler,
(uint32_t)&DMA1_Stream7_IRQHandler,
(uint32_t)&FSMC_IRQHandler,
(uint32_t)&SDIO_IRQHandler,
(uint32_t)&SDIO_IRQHandler,
(uint32_t)&TIM5_IRQHandler,
(uint32_t)&SPI3_IRQHandler,
(uint32_t)&UART4_IRQHandler,
(uint32_t)&UART5_IRQHandler,
(uint32_t)&TIM6_DAC_IRQHandler,
(uint32_t)&TIM7_IRQHandler,
(uint32_t)&DMA2_Stream0_IRQHandler,
(uint32_t)&DMA2_Stream1_IRQHandler,
(uint32_t)&DMA2_Stream2_IRQHandler,
(uint32_t)&DMA2_Stream3_IRQHandler,
(uint32_t)&DMA2_Stream4_IRQHandler,
(uint32_t)&ETH_IRQHandler,
(uint32_t)&ETH_WKUP_IRQHandler,
(uint32_t)&CAN2_TX_IRQHandler,
(uint32_t)&CAN2_RX0_IRQHandler,
(uint32_t)&CAN2_RX1_IRQHandler,
(uint32_t)&CAN2_SCE_IRQHandler,
(uint32_t)&OTG_FS_IRQHandler,
(uint32_t)&DMA2_Stream5_IRQHandler,
(uint32_t)&DMA2_Stream6_IRQHandler,
(uint32_t)&DMA2_Stream7_IRQHandler,
(uint32_t)&USART6_IRQHandler,
(uint32_t)&I2C3_EV_IRQHandler,
(uint32_t)&I2C3_ER_IRQHandler,
(uint32_t)&OTG_HS_EP1_OUT_IRQHandler,
(uint32_t)&OTG_HS_WKUP_IRQHandler,
(uint32_t)&OTG_HS_IRQHandler,
(uint32_t)&DCMI_IRQHandler,
(uint32_t)&CRYP_IRQHandler,
(uint32_t)&HASH_RNG_IRQHandler,
(uint32_t)&HASH_RNG_IRQHandler,
(uint32_t)&FPU_IRQHandler,

};

void Reset_Handler(void)
{
    /* Copy .data section to SRAM*/

    /* Init the .bss section to zero in SRAM */

    /*Init function for standard library */

    /* call main() */
}

void Default_Handler(void)
{
    while (1)
        ;
}

```